

Influence of Prior Acquaintance on the Shared VR Experience

Esen Küçüktütüncü

Ramon Oliva

Mel Slater*

Event Lab, Institute of Neurosciences of the University of Barcelona, Barcelona, Spain

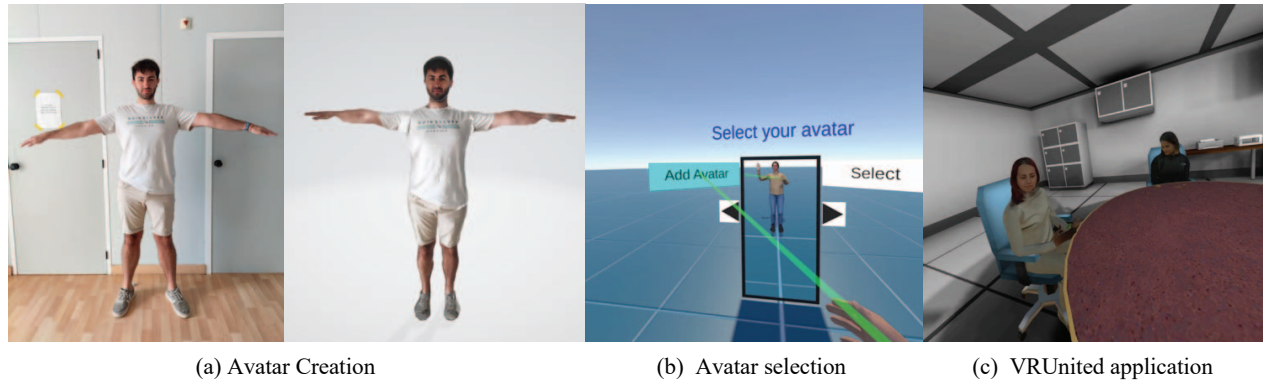


Figure 1: Experiment layout through different perspectives: (a) computer vision pipeline for the avatar creation, (b) choosing your avatar in virtual reality, and (c) virtual meeting room where participants engage with.

ABSTRACT

Exploring social dynamics in virtual reality (VR) at a qualitative level holds great potential for improved applications. Here we examined the influence of prior acquaintance on how people interacted with each other in VR. Groups of 3 or 4 participants, represented by realistic look-alike avatars, engaged in discussions on predefined themes. There were two conditions: (1) groups of individuals with prior connections and (2) strangers. Questionnaire responses revealed that pre-existing acquaintances fostered a stronger sense of copresence, and greater sentiment compared to the strangers group. This insight is crucial for optimizing the design and dynamics of VR interactions.

Keywords: Virtual Reality, shared virtual environments, social identity theory, sentiment analysis, realistic avatars

Index Terms: Human-centered computing—Human-centered computing—Human computer interaction (HCI)—Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Social presence in VR involves a complex interplay of factors, including realistic avatars, synchronous communication, and shared activities. It is crucial for creating a sense of co-presence, making virtual interactions feel more genuine and socially impactful [1]. In their systematic review, Oh et al. [2] explored the determinants of social presence in remote interactions, pinpointing that a combination of high-quality photorealistic representation and consistent behavioral realism significantly contribute to the sense of connection [2]. The proximity between virtual participants emerged as another influential factor, emphasizing that closer virtual distances enhance social presence. Social Identity Theory [3] posits that individuals categorize themselves and others into social groups, influencing intergroup behavior.

When applied in VR, prior acquaintance serves as a basis for a shared identity, promoting a sense of belonging and cooperation within the virtual space. Our scenario, featuring participants resembling their real selves, engaging in full upper-body gestures, and seated close together, aligns with these identified impactful elements.

2 METHODS

For simultaneous presence, the groups met using VRUnited [4], a shared virtual environment application developed using the QuickVR library [5]. We employed an automated system to generate a lifelike 3D avatar of an individual only from a frontal RGB image [6]. This process involves utilizing deep learning techniques to extract pose, shape, and semantic details from the input image. Through this approach, each avatar was successfully generated within approximately 30 minutes.

Once the experimental session started, a moderator also in the VR led a discussion on the following topics:

- (1) Has social media has caused more negative or positive effects on mental health?
- (2) Does personal appearance affect how people respond to us?
- (3) Are techniques such as mindfulness, yoga, meditation etc. really beneficial or are people being deluded?

The questions were designed to be of general interest and were posed by the moderator, all three questions in one session. The discussions typically lasted 20 minutes and, on their conclusion, each participant was asked to write an essay of 150-500 words reflecting their experience. The essay was followed by a Presence and a Social Presence questionnaire.

3 RESULTS

A total of 50 participants were recruited from around the University of Barcelona campus and the city. The participants consisted of 16 groups, 8 for Friends condition, where people had prior acquaintance and were recruited together as a group, and 8 with individuals who were strangers to each other. This Strangers condition had 16 female participants, mean age 24.9 ± 6.84 SD and Friends condition had 17 female participants, with mean age 25.4

* email address: melslater@ub.edu

± 5.87 SD. The questionnaire results showed that for both the ‘feeling of being there’ and copresence, the entire interquartile range for the Friends group was greater than the median of the Strangers group.

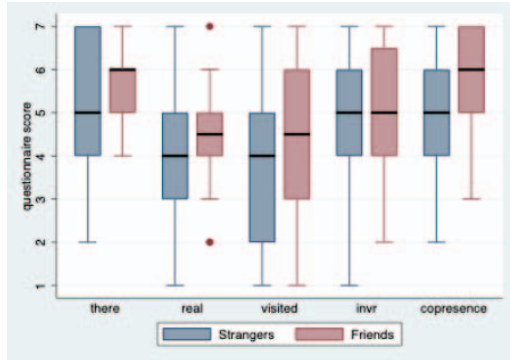


Figure 2: Box plots for the presence related variables, which are a result of Likert scale questions. **there**: the sense of being in a real place. **real**: The extent to which were there times during the experience when the VR was the reality, **visited**: experiencing the VR as images seen, or as somewhere visited. **invr**: the sense of being in the virtual environment, or of being elsewhere. **copresence**: the sense of being in the same space with other people.

We used the Hugging Face system in Python with the BERT [7] model to analyze the essays. This predicts sentiment with a score between 1(lowest) and 5 (highest) sentiment. The mean number of words in essays for the "Strangers" group was 152 with a SD of 51.4, while the "Friends" group had a mean of 178 with a SD of 76.7. The overall mean for both groups combined was 165, with a corresponding standard deviation of 64.8. From distribution of sentiment scores by condition it is clear that the overwhelming sentiment was high for both groups (83% overall had scores of at least 4) (see Table 1).

No. of stars	Strangers	%	Friends	%
1	0	0	0	0
2	1	4	1	4
3	6	23	0	0
4	18	69	20	83
5	1	4	3	13

Table 1: Sentiment analysis results by condition

Although the sentiment is highly positive overall, there is more negative sentiment amongst those in the Strangers group (23% had 3 or less stars compared to 0 in the Friends group). We analyzed all data with Bayesian ordered logistical model for the questionnaire and stars variables and a Bayesian normal regression model for the number of words. We used wide-variance non-informative priors. The resulting posterior probabilities are shown in Table 2. In the model results we found that age and gender (being female) were positively associated with most of the response variables. The responses ‘there’ and ‘invr’, rely on the VR system affording sensorimotor contingencies for perception similar to those in reality [8]. Since this was the same for both groups, we would not expect there to be an important difference between them.

variable	Probability (Friends> Strangers)	Other positively associated variables
copresence	0.934	age
there	0.634	age and being female
sentiment	0.953	being female
virtuality	0.970	age and being female
no. words	0.820	being female

Figure 2: Variables in relations to demographics and conditions, the variable **virtuality** stands for whether the fact that the discussion was virtual did not take away from its quality.

4 DISCUSSION

Our study examined how prior acquaintance affects experiences in VR encounters. Clear differences were found between the groups for copresence and sentiment. While overall sentiment was positive, the Strangers group showed a higher percentage of lower sentiment scores, indicating more negative feelings compared to the Friends group. The observed differences in copresence and sentiment emphasize the importance of considering prior social connections in designing VR applications. This can guide the optimization of VR interaction dynamics, fostering more authentic and socially impactful virtual environments.

ACKNOWLEDGEMENTS

The work is funded by the European Union Horizon 2020 research and innovation programme under grant agreement No. 101017884 - GuestXR project. EK also received support by a grant from Facebook. Facebook did not participate in any way in the study analysis or writing of the results.

REFERENCES

- [1] F. Biocca, C. Harms, and J. K. Burgoon, "Toward a More Robust Theory and Measure of Social Presence: Review and Suggested Criteria," *Presence: Teleoperators and Virtual Environments*, vol. 12, no. 5, pp. 456–480, Oct. 2003, doi: <https://doi.org/10.1162/105474603322761270>.
- [2] C. S. Oh, J. N. Bailenson, and G. F. Welch, "A systematic review of social presence: Definition, antecedents, and implications," *Frontiers in Robotics and AI*, vol. 5, no. 114, Oct. 2018, doi: <https://doi.org/10.3389/frobt.2018.00114>.
- [3] H. Tajfel, J. C. Turner, "An integrative theory of intergroup conflict", *The social psychology of intergroup relations*, pp. 33–47, 1979
- [4] R. Oliva, A. Beacco, J. Gallego, R. G. Abellan and M. Slater, "The Making of a Newspaper Interview in Virtual Reality: Realistic Avatars, Philosophy, and Sushi," in *IEEE Computer Graphics and Applications*, vol. 43, no. 6, pp. 117-125, Nov.-Dec. 2023, doi: 10.1109/MCG.2023.3315761.
- [5] R. Oliva, A. Beacco, X. Navarro, and M. Slater, "QuickVR: A standard library for virtual embodiment in unity," *Frontiers in Virtual Reality*, vol. 3, Sep. 2022, doi: <https://doi.org/10.3389/frvir.2022.937191>.
- [6] A. Beacco, J. Gallego and M. Slater, "Automatic 3D avatar generation from a single RGB frontal image", *Proc. IEEE Conf. Virtual Reality 3D User Interfaces Abstr. Workshops*, pp. 764-765, 2022.
- [7] "nlptown/bert-base-multilingual-uncased-sentiment- Hugging Face," [huggingface.co.https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment](https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment)
- [8] Slater, M., Banakou, D., Beacco, A., Gallego, J., Macia-Varela, F. & Oliva, R. A Separate Reality: An Update on Place Illusion and Plausibility in Virtual Reality. *Front. Virtual Real.* 3, doi:10.3389/frvir.2022.914392 (2022).